

UNIT – 5

(Building Physics/Ergonomics/Comfort)

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Thermal characteristics of a building: Needs of people and the environment

When people are dissatisfied with their thermal environment, not only is it a potential health hazard, it also impacts on their ability to function effectively, their satisfaction at work, the likelihood they will remain a customer, and so on.

BS EN ISO 7730 defines thermal comfort as '...that condition of mind which expresses satisfaction with the thermal environment.', i.e. the condition when someone is not feeling either too hot or too cold.

The human thermal environment is not straight forward and cannot be expressed in degrees. Nor can it be satisfactorily defined by acceptable temperature ranges. It is a personal experience dependent on a great number of criteria and can be different from one person to another within the same space.

For example, a person walking up stairs in a cold environment whilst wearing a coat might feel too hot, whilst someone sat still in a shirt in the same environment might feel too cold.

The Health and Safety Executive (HSE) suggest that an environment can be said to achieve 'reasonable comfort' when at least 80% of its occupants are thermally comfortable. This means that thermal comfort can be assessed by surveying occupants to find out whether they are dissatisfied with their thermal environment.

Contributing factors for thermal comfort

Thermal comfort results from a combination of environmental factors and personal factors:

Environmental factors

Air temperature

The temperature of the air that a person is in contact with, measured by the dry bulb temperature (DBT).

Air velocity

The velocity of the air that a person is in contact with (measured in m/s). The faster the air is moving, the greater the exchange of heat between the person and the air (for example, draughts generally make us feel colder).

Radiant temperature

The temperature of a person's surroundings (including surfaces, heat generating equipment, the sun and the sky). This is generally expressed as mean radiant temperature (MRT, a weighted average of the temperature of the surfaces surrounding a person, which can be approximated by globe thermometer) and any strong mono-directional radiation such as radiation from the sun.

Relative humidity (RH)

The ratio between the actual amount of water vapour in the air and the maximum amount of water vapour that the air can hold at that air temperature, expressed as a percentage. The higher the relative humidity, the more difficult it is to lose heat through the evaporation of sweat.

Personal factors

Clothing

Clothes insulate a person from exchanging heat with the surrounding air and surfaces as well as affecting the loss of heat through the evaporation of sweat. Clothing can be directly controlled by a person (i.e. they can take off or put on a jacket) whereas environmental factors may be beyond their control.

Metabolic heat or level of activity

The heat we produce through physical activity. A stationary person will tend to feel cooler than a person who is exercising.

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Wellbeing and sicknesses

Such as the common cold or flu which affect our ability to maintain a body temperature of 37°C at the core.

Other contributing factors can include; access to food and drink, acclimatisation (this can be more difficult where there is a high outdoor-indoor temperature gradient) and state of health.

In addition, thermal comfort will be affected by whether a thermal environment is uniform or not. For example, draughts and heaters can create a scorched face / frozen back effect and hot feet/cold head and hands effect.

'Thermal alliesthesia' goes beyond this, proposing that the hedonic qualities of the thermal environment (qualities of pleasantness or unpleasantness, or 'the pleasure principle') are determined as much by the general thermal state of the subject as by the environment itself.

In its simplest form, cold stimuli will be perceived as pleasant by someone who is warm, whilst warm stimuli will be experienced as pleasant by someone who is cold. Introducing a spatial component to this, it can for example be pleasurable to wrap cool hands around a warm mug. See Thermal pleasure in the built environment for more information.

Controlling thermal comfort

Thermal comfort can be controlled or adjusted by a number of different measures:

- Environmental monitoring and control (automated or user-controlled systems, active systems such as heating and cooling and passive systems such as shading). NB: User-controlled systems require that users are properly trained.
- Adapting or changing clothing. Businesses can allow people to wear different clothing depending on conditions. They can also provide things like cloak rooms or lockers so that people can change clothes or take off and put down coats. The golden rule is layering, generally 3 layers, and use zips and buttons to regulate temperature.
- Allowing flexible working hours or changing start and finish times.

- Adjusting tasks. For example, allowing breaks or reducing the length of time people are exposed to particular conditions.
- Providing information telling people what sort of conditions to expect so that they can dress and behave appropriately.
- Providing or allowing personal equipment such as desk fans.
- Separating people from sources of discomfort. For example, putting heat generating equipment such as ICT equipment in separate rooms, insulating pipes, preventing draughts and so on. NB: Draughts can be caused by high local surface temperature differences even in a space where there is no air infiltration – for example, a cold down-draught near a window.

Providing protective clothing (PPE Personal Protective Equipment). This should be a last resort option.

Predicting thermal comfort

There are a great number of techniques for estimating likely thermal comfort, including; effective temperature, equivalent temperature, Wet Bulb Globe Temperature (WBGT), resultant temperature and so on, and charts exist showing predicted comfort zones within ranges of conditions.

However, BS EN ISO 7730 and BS EN ISO 10551 suggest thermal comfort can be expressed in terms of Predicted Mean Vote (PMV) and Percentage People Dissatisfied (PPD).

PMV and PPD were developed by Professor Ole Fanger based on research undertaken at Kansas State University and the Technical University of Denmark. Research was carried out to find out if people felt comfortable in different conditions and this was used to develop equations that would predict comfort. The equations take into account; air temperature, mean radiant temperature, air movement, humidity, clothing and activity level.

PMV is an index that predicts the mean vote of a group of people voting on how comfortable they are in an environment. PPD is a function of PMV.

Where non-uniform conditions exist, multiple assessments may be necessary, and in complex environments, Computational Fluid Dynamics (CFD) analysis may be necessary to accurately assess thermal comfort.

NB BREEAM UK New Construction, Non-domestic Buildings (United Kingdom), Technical Manual, SD5078: BREEAM UK New Construction 2018 3.0, published by BRE Global Limited suggests that: 'In BS EN ISO 7730:2005: Ergonomics of the thermal environment. Analytical determination and interpretation of thermal comfort, thermal comfort is defined using the calculation of PMV and PPD indices and local thermal comfort criteria. It is also defined as 'that condition of mind which expresses satisfaction with the thermal environment.' The term 'thermal comfort' describes a person's psychological state of mind and is usually referred to in terms of whether someone is feeling too hot or too cold. Thermal comfort is difficult to define because it needs to account for a range of environmental and personal factors in order to establish what makes people feel comfortable. HSE considers 80% of occupants as a reasonable limit for the minimum number of people who should be thermally comfortable in an environment. The purpose of this issue is to encourage appropriate and robust consideration of thermal comfort issues, and specification of appropriate occupant controls to ensure both maximum flexibility of the space and thermal comfort for the majority of building occupants.'

It also suggests that: 'Thermal comfort analysis tools can be subdivided into a number of methods of increasing complexity. The most complex of these and the one that provides greatest confidence in results is the full dynamic model. This type of model enables annual heating or cooling loads, overheating risks and control strategies to be assessed.'

Regulation

Temperatures in the workplace are governed by the Workplace (Health, Safety and Welfare) Regulations 1992, which oblige employers to provide a reasonable temperature in the workplace.

The Approved Code of Practice (Workplace health, safety and welfare. Workplace (Health, Safety and Welfare) Regulations 1992. Approved Code of Practice) suggests a minimum temperature of 16 degrees Celsius, or 13 degrees Celsius if work involves severe physical effort. However, these are only guidelines.

The Health and Safety Executive (HSE) previously defined thermal comfort in the workplace, as: '...roughly between 13°C (56°F) and 30°C (86°F), with acceptable temperatures for more strenuous work activities concentrated towards the bottom end of the range, and more sedentary activities towards the higher end.'

However, the complexity of thermal comfort means that there is no meaningful maximum guideline temperature, particularly at higher temperatures.

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Factors influencing Thermal Comfort

Air Temperature

It is the temperature of the air surrounding the body. It is usually given in degrees Celsius ($^{\circ}\text{C}$). Right temperature and air freshness are the most important about a building, Freshness of air closely is related to the temperature

Radiant Temperature

Thermal radiation is the heat that emits from a warm object. Radiant heat may be present if there are any heat sources in the environment. Radiant temperature has a greater influence than air temperature on how we lose or gain heat to the environment.

Air Velocity

It describes the speed of air moving across the people and may help cool them if the air is cooler than the environment. Air velocity is an essential factor in thermal comfort for example: still or stagnant air in indoor environments that are artificially heated may cause people to feel airless. It may also lead to a build up odor in moving air in warm or humid conditions which can increase heat loss through convection without any change in air temperature physical activity also increases air movement, so air velocity may be corrected to account for a person's level of physical activity slight air movements in cool or cold environments may be perceived as a draught as people are particularly sensitive to these movements.

Humidity

Relative humidity is the ratio between the actual amount of water vapor in the air and the maximum amount of water vapor that the air can hold at an air temperature. Relative humidity between 40 and 70% does not have a

major impact on thermal comfort. In rooms that are not air conditioned, or where the weather conditions outdoors may influence the indoor thermal environment and relative humidity may be higher than 70%. High humidity environments have a lot of vapor in the air, which inhibits the evaporation of sweat from the skin. In hot environments, humidity is substantial because less perspiration evaporates when humidity is high (80 %+).

The evaporation of sweat is the main method of heat reduction.

Climate responsive design

The climate responsive design refers to the architecture that reflects the particular region-specific weather conditions of the peculiar area. It uses data of weather patterns and factors like sun, wind, rainfall, and humidity. The building structure is built according to the same.

The goal of climate-responsive architecture is to create a **comfortable interior** while reducing the building's reliance on artificial energy. A climate-responsive building design reflects the weather conditions in the precise area where the building is constructed. The design utilizes data on the region's weather patterns and accounts for factors like seasonality, intensity of the sun, wind, rainfall and humidity.

Several elements play a role in limiting a building's energy use based on its site-specific conditions. For example, the building envelope is an important mediator between the indoor and outdoor condition. Envelopes in different climate zones require different assemblies to minimize unwanted energy loss. In the United States, local building codes aim to reduce new buildings' energy use, designating which construction materials and envelope assemblies may be used.

True climate-responsive architecture goes well beyond adherence to code. Smart glazing systems, for example, offer opportunities to further the practice. Electrochromic glass actively controls how much solar light and heat transfer into an interior space. Areas in the West and Southwest United States, for example, experience strong sun and many clear-sky days with bright sunlight. SageGlass minimizes solar heat penetration in summer months to reduce a building's cooling load; conversely, it can be used to maximize solar heat gain during winter months to help reduce the amount of energy needed to heat the building. Though glazing causes more heat transfer than a wall surface, a high-performing system like SageGlass helps climate-

responsive projects effectively manage heat gain while capitalizing on the many benefits that a glazed exterior has on a building's occupants.

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PASSIVE THERMAL CONTROL AND PROTECTION

Without the use of fluid loops or other active parts, passive thermal control regulates the temperatures of a spacecraft and its individual parts. The emphasis is instead on using insulations, careful material selection with consideration for qualities like density, specific heat and conductivity, interface conductance, thermo-optical coatings, and frequently, geometry and spacecraft orientation to regulate the entrance and outflow of heat. The environment must be taken into consideration by passive thermal engineers who use natural heating sources such incoming solar flux, reflected solar flux (albedo), and planetary infrared heating. Radiation is used to cool the surroundings of the space sink, which is very close to absolute zero.

Passive thermal engineers work seamlessly with those focused on active thermal control. The boundary between these two disciplines is often organizational and not technical. There is some overlap between these two focus areas in that passive thermal engineers often rely on some active or semi-active components such as heaters and heat pipes and active thermal engineers rely on environment, thermo-optical properties and other passive tools. Thermal protection focuses primarily on high temperature, high heating type environments often found during ascent and atmospheric re entry. Whereas active and passive thermal practitioners focus on temperatures ranging from hundreds of degrees below zero (F) to, perhaps, a few hundred degrees above zero (F), thermal protection specialists focus on temperatures from many hundreds of degrees (F) to thousands of degrees (F). Through use of reusable thermal protection materials such as silica tiles and quartz blankets, high temperature seals as well as ablative materials, thermal protection engineers ensure the spacecraft structure, key spacecraft components and, of course, crew are protected from the harsh ascent and entry heating environments. Of special interest is the area of hot structures, such as the reinforced carbon-carbon found on the Space Shuttle Orbiter. This material is not an insulator. Rather, it is material that is designed to survive temperatures in excess of 3000 F, sustain aerodynamic load and be used over and over again

Thermal balance

Thermal balance definition. Thermal balance occurs **when the sum of all the different types of heat flow into and out of a building is zero**. That is, the building is losing as much heat as it gains so it can be said to be in equilibrium.

Active heating/cooling

Active solar heating systems use **solar energy to heat** a fluid -- either liquid or air -- and then transfer the solar heat directly to the interior space or to a storage system for later use. Both liquid and air systems can supplement forced air systems.

Lighting aspects of a building

Physics of light

Light is electromagnetic radiation that can be detected by the human eye. Electromagnetic radiation occurs over an extremely wide range of wavelengths, from gamma rays with wavelengths less than about 1×10^{-11} metres to radio waves measured in metres.

Photometry

Photometry is the science of measuring visible light as perceived by the human eye. We use it mainly in the context of lighting technology whether in architectural, photographic, audiovisual, or other similar settings.

The measuring unit of illuminance is the lux, or foot-candle; it indicates the amount of illumination a given surface unit receives.

We use two devices to measure photometric data, namely the number of lumens produced by a light source:

- **The integrating sphere** is used to measure all electrical features of a luminaire as well as the chromatic properties of a light source, particularly the luminous flux, or lumens
- **The goniophotometer** measures the spatial and angular distribution of a light source

The 5 Basic Metric Values

- **Luminous Intensity (candela):** It is the amount of light dispersed in a given direction. It is calculated in large part by guiding parts of the lighting flux such as reflectors. It is illustrated by the photometric curve.
- **Luminous flux (lumen):** It is the amount of light produced by a light source. As for light efficacy, it is the result of luminous flux divided by the amount of electrical energy consumed (lumen/watts). It reflects the profitability of a light source.
- **Luminous existence (candela/ft²):** It represents the spatial density of the luminous flux starting from a specific location.
- **Luminous (candela/m²):** Illuminance is the sole photometric magnitude that the human eye can determine. It represents the amount of luminous intensity (candela) that an object will reflect or produce per square meter.
- **Illuminance (candela/m²):** Illuminance is a measurement of the amount of luminosity that light projects onto a given surface. More specifically, it is the density of the luminous flux (lumens) on a surface. It tends to diminish as the distance between the surface and the light source increases (Inverse Square Law).

Concept of sun light and ventilation

Infections caught in buildings are a major global cause of sickness and mortality. Understanding how infections spread is pivotal to public health yet current knowledge of indoor transmission remains poor.

Sunlight and resistance to infection

Receiving direct sunshine, even from behind glass, may improve health and infection resistance. Seasonal and severe depression are just two of the psychiatric illnesses that are treated by bright-light treatment. Hospital patients with mental illnesses and other disorders heal more quickly in sunlit wards, and sunlight has a good impact on length of stay, mortality rate, perceived stress, and pain. Beyond reducing the symptoms of sadness, bright light has further advantages since it lowers the rate of cognitive deterioration in dementia patients. Without correct time cues from the sun or other sources, fundamental biological rhythms may fall out of sync, leading to a variety of health issues, such as depression, diabetes, obesity, breast and prostate cancer, and other conditions. Regulating biological rhythms in

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patients may enhance immunolog

ical activity but it is not known whether habitual exposure to sunlight behind glass would help patients withstand infection

LIGHTING AND VENTILATION OF ROOMS

a) Rooms: Every habitable room which should have for the admission of air and light, one or more apertures such as windows and fanlights, opening directly to the external air or into an open verandah and of an aggregate area, inclusive of frames, of not less than

i. One-tenth of the floor area excluding doors for dry hot climate.

ii. One-sixth of the floor area excluding doors for wet/hot climate.

No portion of a room should be assumed as lighted if it is more than 7.5m away from the door or window which is taken for calculation as ventilating that portion.

Cross-ventilation by means of windows and ventilators or both shall be effected in at least living room of tenement either by means of windows in opposite walls or if this is not possible or advisable, then at least in the adjoining walls.

Vision and colours

Color vision, a feature of visual perception, is an ability to perceive differences between light composed of different wavelengths (i.e., different spectral power distributions) independently of light intensity. Color perception is a part of the larger visual system and is mediated by a complex process between neurons that begins with differential stimulation of different types of photoreceptors by light entering the eye. Those photoreceptors then emit outputs that are propagated through many layers of neurons and then ultimately to the brain. Color vision is found in many animals and is mediated by similar underlying mechanisms with common types of biological molecules and a complex history of evolution in different animal taxa. In primates, color vision may have evolved under selective pressure for a variety of visual tasks including the foraging for nutritious young leaves, ripe fruit, and flowers, as well as detecting predator camouflage and emotional states in other primates.

Visual comfort

Visual comfort and daylighting expert, Laura Thuillier, talks about the evolving definition and measurement of visual comfort and how we can design buildings that meet personal preferences and individual needs.

How is visual comfort measured?

Visual comfort is usually defined through a set of criteria based on the level of light in a room, the balance of contrasts, the colour 'temperature' and the absence or presence of glare.

There are actually many different definitions of visual comfort and what you should measure as part of it.

At Saint-Gobain, we've developed our definition of visual comfort based on what we believe is the best combination of factors to optimise well-being and the feeling of comfort. This includes the quantity and quality of light into a building, quality and access to views from inside the building and the quality of the surrounding space.

This said, visual comfort is very much an emerging field of study and new metrics are proposed almost every day.

The most widely used and accepted metrics tend to focus on:

- **The measure or the quantity of natural light over the year:** the potential of a building and its location to provide enough daylight to occupants. This is typically explored through daylight autonomy or useful daylight illuminance
- **The distribution of light as perceived by the eye:** whether a space will be too bright or too highly contrasting
- **Others:** there are other concepts that don't have a metric but on which academics widely agree. The quality of the light (through its spectral composition) is one example. The quality of a view is also a consideration but is hard to quantify; however, the first methodologies are starting to appear

Electric lighting

A gadget that uses electric power to produce visible light is called an electric light. Light bulbs, such as the incandescent light bulb, are widely used to refer to lamps. Typically, lamps have a base composed of ceramic, metal, glass, or plastic that keeps the lamp in the light fixture's socket.

Acoustic aspects of a building- Noise insulation

Building acoustics is the science of controlling noise in buildings. This includes the minimisation of noise transmission from one space to another and the control of the characteristics of sound within spaces themselves.

Building acoustics are an important consideration in the design, operation and construction of most buildings, and can have a significant impact on health and wellbeing, communication and productivity. They can be particularly significant in spaces such as concert halls, recording studios, lecture theatres, and so on, where the quality of sound and its intelligibility are very important.

Building acoustics can be influenced by:

- The geometry and volume of a space.
- The sound absorption, transmission and reflection characteristics of surfaces enclosing the space and within the space.
- The sound absorption, transmission and reflection characteristics of materials separating spaces.
- The generation of sound inside or outside the space.
- Airborne sound transmission.
- Impact noise.

Characteristics of sound

Sound intensity is measured in Decibels (dB). This is a logarithmic scale in which an increase of 10 dB gives an apparent doubling of loudness.

Sound pitch is measured in Hertz (Hz), the standard unit for the measurement for frequency. The audible range of sound for humans is typically from 20 Hz to 20,000 Hz, although, through ageing and exposure to loud sounds the upper limit will generally decrease.

As well as intensity and frequency, sound also transmits information. For example, music or speech, transmit information which people may perceive differently from other sounds.

Reverberation time

The 'reverberation time' of a space changes the way the space 'sounds' and can affect the intelligibility acoustic information. A high reverberation time can make a room sound muffled, loud and noisy. Rooms designed for speech typically have a low reverberation time, whereas a higher reverberation time can add depth, richness and warmth to music.

The reverberation time of a room is defined as the time it takes for sound to decay by 60 dB after an abrupt termination. It is linked to the total quantity of soft treatments and the volume of the room.

Sound absorption

Sound absorption is the loss of sound energy when sound waves come into contact with an absorbent material such as ceilings, walls, floors and other objects, as a result of which, the sound is not reflected back into the space. Acoustic absorption can be used to reduce reverberation times.

Absorbent materials are sometimes categorised from A to E, where A is highly absorbent and E is almost fully reflective.

Sound absorbers can be divided into three main categories:

- Porous absorbents, such as fibrous materials or open-celled foam.
- Resonance absorbents, which consist of a mechanical or acoustic oscillation system, such as membrane absorbers.
- Single absorbers such as tables, chairs or other objects.

Sound insulation

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Sound transmission paths can be interrupted by sound insulation and by blocking air paths. The sound insulation of a single leaf of a material is governed by its mass, stiffening and damping.

The sound insulation across a good conventional, lightweight, office to office construction is typically in the order of 45 dB Dw. This means that if the sound level in the source room is around 65 dB (a typical level for speech), the sound level in the adjacent room, the receiver room, will be approximately 20 dB (barely audible).

If sound levels are increased in the source room however, to 75 dB (raised voice), sound levels within the adjacent room will also increase to around 30 dB (audible). Sound insulation therefore describes the level of sound lost across a partition and not the level of sound within an adjacent room.

Dw represents the sound insulation between rooms on-site. Rw represents the lab tested sound insulation of an element making up a partition wall/floor type. Standards achieved in labs may not be possible on site because of the quality of workmanship and due to sound 'flanking' acoustic elements, that is, travelling around them through an easier path, rather than only directly through them as under lab conditions.

The building regulations part E sets minimum standards for design and construction in relation to the resistance to the passage of sound.

Noise nuisance

Building acoustics can help to mitigate the effects of noise disturbance which can have negative effects on health, wellbeing and general quality of life.

The Noise Policy Statement for England (NPSE) defines noise pollution as:

- Environmental noise – which includes noise from transportation sources.
- Neighbour noise – which includes noise from inside and outside buildings.
- Neighbourhood noise – which includes noise arising from industrial and entertainment premises, trade and businesses, construction sites and noise in the street.

This can be an important consideration for the location, design and construction of new developments.

Room acoustics

room acoustics is defined as how sound behaves in an enclosed space in terms of the reverberation time (or degree of echo), overall noise levels and speech intelligibility. Room acoustics are influenced by room geometry and distribution of acoustic absorption either through the general room finishes or through the introduction of sound absorbing products.'

Basics of Sound, the Ear, and Hearing

In this chapter we review basic information about sound and about how the human auditory system performs the process called hearing. We describe some fundamental auditory functions that humans perform in their everyday lives, as well as some environmental variables that may complicate the hearing task. We also discuss the types of hearing loss or disorder that can occur and their causes.

INTRODUCTION TO SOUND

Hearing makes it possible to communicate via sound since it enables one to identify and recognise items in the world based on the sounds they create. A sound-transmitting medium, such as air, experiences pressure fluctuations as a result of an object's vibration, which produces sound. A pressure wave is sent from the source of vibration outward. The pressure wave can transfer vibration to another object when it comes into contact with it, and the pressure wave will then move through the medium of the other object. The sound wave may also bounce off the item or diffract in its vicinity. As a result, a sound wave that is emitted from a vibrating item might reach a listener's eardrum, causing the eardrum to vibrate and starting the hearing process.

Sound waves can be mathematically described in two ways, that is, in two domains. In the time domain, sound is described as a sequence of pressure changes (oscillations) that occur over time. In other words, the time-domain description of a sound wave specifies how the sound pressure increases and decreases over time. In the frequency domain, the spectrum defines

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sound in terms of the tonal components that make up the sound. A tonal sound has a time-domain description in which sound pressure changes as a regular (sinusoidal) function of time. If one knows the tonal components of sound as defined in the frequency domain, one can calculate the time-domain description of the sound. Using the same analytic tools, the frequency domain representation of a sound can also be calculated from the time-domain description. Thus, the time and frequency domain descriptions of sound are two different ways of measuring the same thing (i.e., the time and frequency domains are functional equivalents). Thus, one can describe sound as temporal fluctuations in pressure, or one can describe sounds in terms of the frequency components that compose the sound.

Tonal sounds have received a lot of attention in the study of hearing, mostly because they serve as the foundation for the frequency domain description of sound. On the other hand, common sounds are complex sounds made up of numerous tonal frequency components. Noise is a typical complicated sound that is employed in hearing research. All potential frequency components are present in noise, and its magnitude changes randomly over time. If all of the frequency components are present and are all at the same average sound level, a noise is considered to be "white noise."

A sound waveform has three basic physical attributes: frequency, amplitude, and temporal variation. Frequency refers to the number of times per second that the vibratory pattern (in the time domain) oscillates. Amplitude refers to sound pressure. There are many aspects to the temporal variation of sound, such as sound duration. Sound pressure is proportional to sound intensity (in units of power or energy), so sound magnitude can be measured in units of pressure, power, and energy. The common measure of sound level is the decibel (dB), in which the decibel is the logarithm of the ratio of two sound intensities or two sound pressures. Frequency is measured in units of hertz (Hz), cycles per second. Measures of time are expressed in various temporal units or can be translated into phase measured in angular degrees. Below are some definitions of terms and measures used to describe sound.

- Sound pressure (p): sound pressure is equal to the force (F) produced by the vibrating object divided by the area (Ar) over which that force is being applied: $p = F/A_r$.
- DekaPascals or daPa; the Système International unit of pressure. One daPa = 100 dynes per cm^2 , and one atmosphere = 10132.5 daPa.

- Sound intensity (I): sound intensity is a measure of power. Sound intensity equals sound pressure squared divided by the density (p_o) of the sound-transmitting medium (e.g., air) times the speed of sound (c): $I = p^2/p_o c$. Energy is a measure of the ability to do work and is equal to power times the duration of the sound, or $E = PT$, where P is power and T is time (duration) in seconds.
- Decibel (dB): $dB = 10 \cdot \log_{10}(I/I_{ref})$ or $20 \cdot \log_{10}(p/p_{ref})$, where I is sound intensity, p is sound pressure, ref is a referent intensity or pressure, and $\log_{10}$ is the logarithm to the base 10. When p_{ref} is 20 micropascals, then the decibel measure is expressed as dB SPL (sound pressure level).
- Hertz (Hz): hertz is the measure of vibratory frequency in which “n” cycles per second of periodic oscillation is “n” Hz.
- Phase (angular degrees): one cycle of a periodic change in sound pressure can be expressed in terms of completing the 360 degrees of a circle. Thus, half a cycle is 180 degrees, and so on. Thus, time (t) within a cycle can be expressed in terms of phase (θ , expressed in degrees), $\theta = 360^\circ(t)(f)$, where f = frequency in Hz, and t = time in seconds.
- Tone (a simple sound): a tone is a sound whose amplitude changes as a sinusoidal function of time: $A \sin(2 \pi f t + \theta)$, where $\sin$ is the trigonometric sin function, θ = peak amplitude, f = frequency in Hz, t = time in seconds, and θ = starting phase in degrees.
- Complex sound: any sound that contains more than one frequency component.
- Spectrum: the description of the frequency components of sound; amplitude spectrum describes the amplitude of each frequency component; phase spectrum describes the phase of each frequency component.
- Noise: a complex sound that contains all frequency components, and whose instantaneous amplitude varies randomly.
- White noise: a noise in which all of the frequency components have the same average level.

The term “noise” can refer to any sound that may be unwanted or may interfere with the detection of a target or signal sound. In some contexts, a speech sound may be the signal or target sound, and another speech sound or a mixture of other speech sounds may be presented

as a “noise” to interfere with the auditory processing of the target speech sound. Often a mixture of speech sounds is referred to as “speech babble

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